UPDATE SYSTEM

```
sudo apt update && sudo apt full-upgrade -y
```

INSTALL PRE REQUISITIES

1. Install Apache
    a. sudo apt install apache2 apache2-doc apache2-utils

2. Setup PHPMyAdmin

    a. Setup something like PHPMyAdmin if you want a graphical user interface rather than the command line. It does make managing a database slightly easier.

    For RPi 4:

    ```
    sudo apt-get install phpmyadmin
    ```

    During the installation, phpMyAdmin will prompt for configuring it. In the first prompt select ‘Apache2’ as the webserver. In the second prompt it will asks if I want it to create a database automatically, I choose ‘no‘ and continue with the installation. I’m choosing ‘no’ in the 2nd step because I’ll setup my custom database for Django instead.

    b. systemctl restart apache2
    systemctl reload apache2

3. Configuring Apache for PHPMyAdmin

    a. To get started, we need to edit the “Apache2.conf” file, we can do this by entering the following into the terminal.

    sudo nano /etc/apache2/apache2.conf

    b. Now we need to add the following line to the bottom of this file.
    This line will include PHPMyAdmin’s configuration and allow it to be loaded in and listened to by Apache.

    Include /etc/phpmyadmin/apache.conf
    sudo systemctl reload apache2

4. Install MariaDB

For RPi 4:

```
sudo apt install mariadb-server mariadb-client

sudo mysql -u root -p

use mysql;

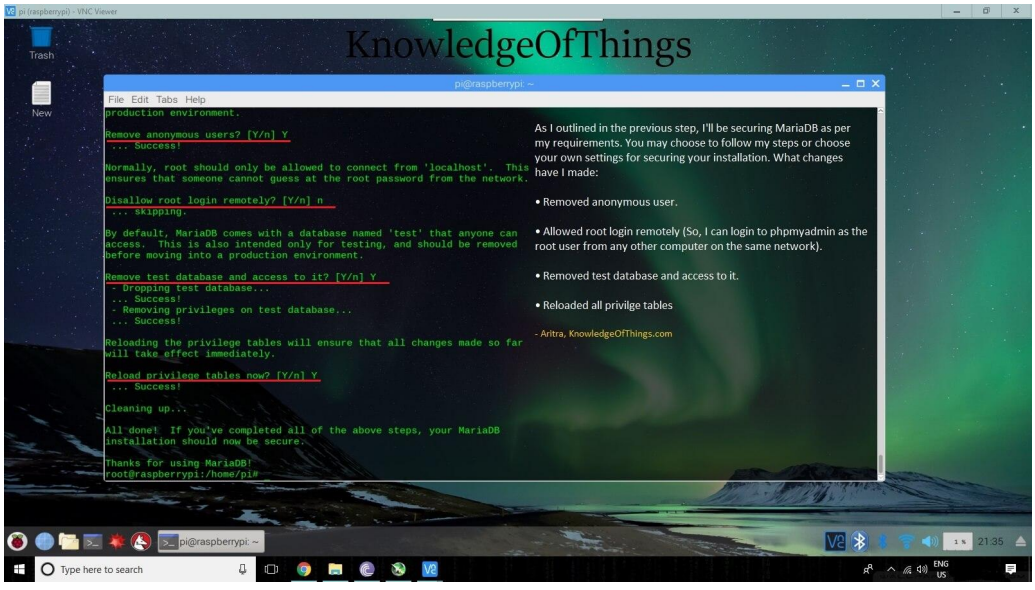
update user set password=PASSWORD("abc123"), plugin = ' ' where User='root';

flush privileges;

\q
```

a. Secure my MariaDB server so that it can be used in production environment.
mysql_secure_installation

- Remove anonymous user
- Allowed root login remotely so that i can login to phpmyadmin as the root user using another device on the same network
- Removed test database
- Reloaded all privilege tables



5. Try accessing phpmyadmin localhost/phpmyadmin

    a. This will open the administrative page and I can login with my administrative user ‘root’ and the password I set previously while configuring MariaDB.

    b. If not found 404

        sudo ln -s /etc/phpmyadmin/apache.conf /etc/apache2/conf-available/phpmyadmin.conf

        sudo a2enconf phpmyadmin.conf

        sudo systemctl reload apache2

SETUP MARIADB TO WORK WITH DJANGO

1. sudo apt-get install python3-pip libapache2-mod-wsgi-py3

2. Create admin user
    a. ./manage.py createsuperuser
3. Run server
    a. ./manage.py runserver 0.0.0.0:8000

4. The first step in setting up MariaDB for Django is to create a database which Django can use. To do so, first login to mysql prompt as the root user (as you have created earlier). Then, create a database by a name you want. Next, create and assign a user to the database. Once done, flush all other privileges to the database.

```
mysql -u root -p   [Hit “Enter” key and provide the password to Login as root user]

CREATE DATABASE thesisDatabase;

CREATE USER 'danedb'@'localhost' IDENTIFIED BY 'abc123';

 GRANT ALL PRIVILEGES ON thesisDatabase.* TO 'danedb'@'localhost';


FLUSH PRIVILEGES;

 \q
```

5. Tell Django to use the newly created database.
    a. settings.py

```
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.mysql',
        'NAME': 'thesisDatabase',
        'USER': 'danedb',
        'PASSWORD': 'abc123',
        'HOST': 'localhost',
        'PORT': '',
    }
}
```

6. Install certain packages that are required to create python apps, and be able to use the MariaDB server system-wide.

    sudo apt-get install python-pip python-dev mariadb-server libmariadbclient-dev libssl-dev

7. Now, after installing the MariaDB-server system-wide, I go on to install the MariaDB client. Before doing that, I need to activate the virtual environment and then install the MariaDB-client. To do these steps, type the following commands:# source /home/pi/Django/DjangoEnv/bin/activate

    pip3 install django mysqlclient

    If cant install this, https://github.com/facebook/prophet/issues/418

8. Save changes

    python3 manage.py makemigrations
    python3 manage.py migrate
    python3 manage.py runserver 0.0.0.0:8000

9. Login to phpmyadmin and check whether the newly created database can be managed with the GUI.